Project: Worked examples
Title: Bolts and fillet welds (EN 1993-1-8)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the design resistance of one bolt in a plate, block tearing of the bolt group, the spacing limits, and a fillet weld by the simplified and directional methods, each against its design force. Loading is static.

WHAT IT CALCULATES	RELATION
Bolt strengths	f_{ub} = class × 100 MPa; $A_s = \pi/4\cdot((d_2 + d_3)/2)^2$, $d_2 = d - 0.6495P$, $d_3 = d - 1.2269P$
Shear (cl. 3.6, Table 3.4)	$F_{v,Rd} = k_{v12}\cdot\alpha_v\cdot f_{ub}\cdot A\cdot n_s/\gamma_{M2}$; α_v = 0.6 (4.6, 5.6, 8.8 or shank in the plane), 0.5 (4.8, 5.8, 6.8, 10.9 with threads in the plane); k_{v12} = 0.85 for an M12 bolt of class 4.8, 5.8, 6.8, 8.8 or 10.9 in its normal clearance hole $d_o = d + 2$ (Table 3.4, note on M12 bolts), else 1
Bolt bearing resistance	$F_{b,Rd} = k_h\cdot\min(k_1\cdot\alpha_b\cdot f_u\cdot d\cdot t/\gamma_{M2}$, [single lap, one bolt row: $1.5\cdot f_u\cdot d\cdot t/\gamma_{M2}$]); $\alpha_b = \min(\alpha_d, f_{ub}/f_u, 1)$, α_d = $e_1/3d_o$ (end) or $p_1/3d_o - 1/4$ (inner); $k_1 = \min(2.8e_2/d_o - 1.7, 1.4p_2/d_o - 1.7, 2.5)$; k_h = 1 (normal holes), 0.8 (oversize), 0.6 (slot across the load)
Bolt tension resistance	$F_{t,Rd} = k_2\cdot f_{ub}\cdot A_s/\gamma_{M2}$, k_2 = 0.9; punching $B_{p,Rd} = 0.6\pi\cdot d_m\cdot t\cdot f_u/\gamma_{M2}$, $d_m = (s + e)/2$ from the across-flats s and the across-corners e of the smaller of head and nut, or d_m = s when e is not entered (conservative); combined $F_v/F_{v,Rd} + F_t/(1.4F_{t,Rd}) \le 1$
Block tearing (cl. 3.10.2)	$V_{eff,1,Rd} = f_u\cdot A_{nt}/\gamma_{M2} + f_y\cdot A_{nv}/(\sqrt{3}\cdot\gamma_{M0})$ for a group loaded concentrically; $V_{eff,2,Rd} = 0.5\cdot f_u\cdot A_{nt}/\gamma_{M2} + f_y\cdot A_{nv}/(\sqrt{3}\cdot\gamma_{M0})$ for an eccentrically loaded group
Weld, simplified (cl. 4.5.3.3)	$f_{vw,d} = f_u/(\sqrt{3}\cdot\beta_w\cdot\gamma_{M2})$, $F_{w,Rd} = f_{vw,d}\cdot a$; directional (cl. 4.5.3.2): $\sqrt{(\sigma^2 + 3(\tau^2 + \tau^2))} \le f_u/(\beta_w\cdot\gamma_{M2})$, $\sigma \le 0.9f_u/\gamma_{M2}$

Basis: Positioning of holes for bolts[1](cl. 3.5, Table 3.3); Design resistance of individual fasteners[1](cl. 3.6.1, Table 3.4); Block tearing[1](cl. 3.10.2); Fillet welds: directional and simplified methods; correlation factor[1](cl. 4.5.3.2, 4.5.3.3, Table 4.1); Nominal stress area (formula)[2](cl. 9.1).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
grade	Bolt property class	8.8		drawing: SPL-001 rev B	any
size	Bolt size	M20		drawing: SPL-001 rev B	any
steel	Plate steel grade (β_w)	S275		drawing: SPL-001 rev B	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$hole_{type}$	Hole type: oversize holes reduce the bearing resistance by 0.8, a slot across the load by 0.6 (a d_o above the normal clearance is treated as oversize)	Normal clearance		default	any
$block_{load}$	Block tearing loading: concentric ($V_{eff,1}$) or eccentric ($V_{eff,2}$, the tension term halved)	Concentric loading (V_eff,1)		default	any
$single_{lap}$	Single-lap joint with only one bolt row (limits the bearing resistance to $1.5 \cdot f_u \cdot d \cdot t / \gamma_{M2}$, cl. 3.6.1)	no		default	any
threads	Threads in the shear plane	yes		assumption: Threads may lie in the shear planes (short bolts).	any
n_s	Shear planes	2		drawing: SPL-001 rev B	1 … 2
t_p	Plate thickness	10	mm	drawing: SPL-001 rev B	4 … 40 mm
e_1	End distance (load direction)	40	mm	drawing: SPL-001 rev B	15 … 250 mm
e_2	Edge distance (across the load)	35	mm	drawing: SPL-001 rev B	15 … 250 mm
p_1	Pitch in the load direction (optional; blank = one row)	70	mm	drawing: SPL-001 rev B	30 … 300 mm
f_y	Plate yield strength	275	MPa	spec: EN 10025-2 S275, t ≤ 16 mm	235 … 440 MPa
f_u	Plate ultimate tensile strength	410	MPa	spec: EN 10025-2 S275, 3 ≤ t ≤ 100 mm	340 … 700 MPa
$F_{v,Ed}$	Shear per bolt	55	kN	assumption: Governing bolt from the bolt-group calculation (M3).	0 … 1000 kN
$F_{t,Ed}$	Tension per bolt	0	kN	assumption: No tension.	0 … 1000 kN
γ_{M2}	Partial factor for bolt and weld resistance	1.25		spec: UK NA to BS EN 1993-1-8, NA.2.3	1.1 … 1.4
γ_{M0}	Partial factor for cross-section resistance	1		spec: UK NA to BS EN 1993-1-1, NA.2.15	1 … 1.1
k_{e1}	Minimum e_1/d_o	1.2		spec: EN 1993-1-8 Table 3.3	1 … 2.5

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
k_{e2}	Minimum e_2/d_o	1.2		spec: EN 1993-1-8 Table 3.3	1 … 2.5
k_{p1}	Minimum p_1/d_o	2.2		spec: EN 1993-1-8 Table 3.3	2 … 3.5
k_{p2}	Minimum p_2/d_o	2.4		spec: EN 1993-1-8 Table 3.3	2 … 3.5

3. Assumptions

A1. Loading is static. (Scope of this sheet.)
A2. Bolts are category A in shear (bearing type) and category D in tension (non-preloaded). (Scope of this sheet.)
A3. The joint has no slip-resistant bolts and no moment end plates. (Scope of this sheet.)
A4. Bearing is taken at the worst of an end bolt and an inner bolt. (Conservative for every bolt in the group.)
A5. Bearing across the load is taken at an edge bolt, with k_1 from e_2 and p_2. (Conservative for every bolt in the group.)
A6. Oversize holes reduce the bearing resistance by 0.8, and a slot across the load by 0.6. (BS EN 1993-1-8:2005 cl. 3.6.1, Table 3.4.)
A7. A hole above the normal clearance, with the hole type left at normal, is treated as oversize. (BS EN 1993-1-8:2005 cl. 3.6.1, Table 3.4.)
A8. A 12 mm bolt in its normal clearance hole takes 0.85 on its shear resistance, except classes 4.6 and 5.6. (BS EN 1993-1-8:2005 Table 3.4, the note on 12 mm and 14 mm bolts in 2 mm clearance holes.)
A9. The single-lap cap on the bearing resistance applies only when the single-lap switch is on. (BS EN 1993-1-8:2005 cl. 3.6.1; a warning says when it would bind.)
A10. Spacing minima and γ_M factors are entered from a National Annex record. (A small set of factors only; no table from a standard is shipped.)
A11. The defaults for the spacing minima and γ_M factors are the recommended values of BS EN 1993-1-8. (A small set of factors only; no table from a standard is shipped.)
A12. Group forces and weld forces per length come from the bolt group and weld group sheets by link. (Linked from the bolt-group and weld-group calculations.)
A13. threads (Threads in the shear plane) is taken as yes. (Threads may lie in the shear planes (short bolts).)
A14. $F_{v,Ed}$ (Shear per bolt) is taken as 55 kN. (Governing bolt from the bolt-group calculation (M3).)
A15. $F_{t,Ed}$ (Tension per bolt) is taken as 0 kN. (No tension.)

4. Working

[20] f_{ub} = class first figure × 100 [1](Table 3.1)
= class 8.8
= 800 MPa
Bolt ultimate tensile strength

[21] f_{yb} = f_{ub} × second figure / 10 [1](Table 3.1)
= class 8.8
= 640 MPa
Bolt yield strength

[22] A_s = π/4·((d_e + d_s)/2)², d_e = d − 0.6495P, d_s = d − 1.2269P [2](cl. 9.1)
= d = 20, P = 2.5 mm
= 244.79 mm²
Tensile stress area

[23] d_0 = d + 2 (≤ M24), d + 3 (≥ M27) unless entered
= 22 mm
= 22 mm
Hole diameter
[24] α_v = α_v (threads in the shear plane) [1](Table 3.4)
= class 8.8
= 0.6
Shear factor
[25] $F_{v,Rd}$ = $\alpha_v \cdot f_{ub} \cdot A \cdot n_s / \gamma_{M2}$ [1](Table 3.4)
= 0.6 × 800 × 244.79 × 2 / 1.25
= 188 kN
Shear resistance per bolt
[26] α_b = min(α_d, f_{ub}/f_u, 1), α_d = min($e_1/3d_0$, $p_1/3d_0$ - 1/4) [1](Table 3.4)
= min(0.60606, 0.81061, 1.9512, 1)
= 0.60606
Bearing factor α_b
[27] k_1 = min($2.8e_2/d_0$ - 1.7, $1.4p_2/d_0$ - 1.7, 2.5) [1](Table 3.4)
= e_2 = 35, p_2 = –, d_0 = 22
= 2.5
Bearing factor k_1
[29] $F_{b,Rd}$ = $k_1 \cdot \alpha_b \cdot f_u \cdot d \cdot t / \gamma_{M2}$ [1](cl. 3.6.1, Table 3.4)
= 2.5 × 0.60606 × 410 × 20 × 10 / 1.25
= 99.394 kN
Bearing resistance per bolt
[30] $F_{t,Rd}$ = $k_2 \cdot f_{ub} \cdot A_s / \gamma_{M2}$, k_2 = 0.9 [1](Table 3.4)
= 0.9 × 800 × 244.79 / 1.25
= 141 kN
Tension resistance per bolt
[31] $B_{p,Rd}$ = $0.6\pi \cdot d_m \cdot t \cdot f_u / \gamma_{M2}$, d_m = s (e not entered) [1](Table 3.4)
= 0.6π × 30 × 10 × 410 / 1.25
= 185.48 kN
Punching shear resistance
[32] u_v = $F_{v,Ed}$ / $F_{v,Rd}$
= 55 kN / 188 kN
= 0.29255
Shear utilisation
[33] u_b = $F_{v,Ed}$ / $F_{b,Rd}$
= 55 kN / 99.394 kN
= 0.55335
Bearing utilisation
[34] u_t = $F_{t,Ed}$ / min($F_{t,Rd}$, $B_{p,Rd}$)
= 0 kN / min(141 kN, 185.48 kN)
= 0
Tension utilisation (tension or punching)
[35] u_{vt} = $F_{v,Ed}$ / $F_{v,Rd}$ + $F_{t,Ed}$ / (1.4 * $F_{t,Rd}$) [1](Table 3.4)
= 55 kN / 188 kN + 0 kN / (1.4 × 141 kN)
= 0.29255
Combined shear and tension
[37] u_{limit} = limit
= the utilisation limit
= 1
Utilisation limit

[38] $F_{t,lim} = \min(F_{t,Rd}, B_{p,Rd})$
$= \min(141\ kN, 185.48\ kN)$
$= 141\ kN$
Governing tension or punching resistance

5. Results

SYMBOL	VALUE	DESCRIPTION
$F_{v,Rd}$	188.00 kN	Shear resistance
$F_{b,Rd}$	99.39 kN	Bearing resistance
$F_{t,Rd}$	141.00 kN	Tension resistance
A_s	244.8 mm²	Stress area
f_{ub}	800 MPa	Bolt ultimate tensile strength
f_{yb}	640 MPa	Bolt yield strength
α_v	0.6	Shear factor
α_b	0.60606	Bearing factor
k_1	2.5	Bearing factor k_1
$B_{p,Rd}$	185.48 kN	Punching resistance
governing	"bearing (0.553)"	Governing bolt mode
u_v	0.29	Shear utilisation
u_b	0.55	Bearing utilisation (governing)
u_t	0.00	Tension utilisation
u_{vt}	0.29	Combined utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Shear force against bolt bearing resistance	55 kN	99.39 kN	0.55	[1](Table 3.4)
PASS	Shear force against bolt shear resistance	55 kN	188.00 kN	0.29	[1](Table 3.4)
PASS	Combined shear and tension against its limit	0.29255	1	0.29	[1](Table 3.4)
PASS	Tension force against bolt tension or punching resistance	0 kN	141.00 kN	0.00	[1](Table 3.4)
PASS	Spacing and edge distances against the Table 3.3 minima	e_1 40, e_2 35, p_1 70, p_2 – mm	≥ 26.4, 26.4, 48.4, 52.8 mm	–	[1](Table 3.3)

7. Applicability

WARNING records: tabulated values not taken from a reference record are γ_{M2}, γ_{M0}, k_{e1}, k_{e2}, k_{p1}, k_{p2} (EN 1993-1-8 National Annex factors); f_y, f_u (Plate strengths). Enter each as one so its source, edition and licence travel with it. [record]

Defaults used (inputs left empty):
$hole_{type}$: not entered, so the default of 'Normal clearance' is used.
$block_{load}$: not entered, so the default of 'Concentric loading ($V_{eff,1}$)' is used.

8. Verification

```
Template struct.connection-ec3@1.9.0 Content hash 9d487fad84a4c3c8 Engine calc-engine/1.0.0
Known-answer cases passed 9/9 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-8: Design of Joints. BS EN 1993-1-8:2005. BSI; 2005.
2. International Organization for Standardization. Mechanical Properties of Fasteners Made of Carbon Steel and Alloy Steel—Part 1: Bolts, Screws and Studs With Specified Property Classes—Coarse Thread and Fine Pitch Thread. ISO 898-1:2013. ISO; 2013.